

Transformer and Attention

Monday, October 6, 2025 7:55 PM

Quiz on Friday

Next week LLMs

What does gradient descent do?

Calculating weights to minimize loss

Why don't we just use more and more layers of a neural net to compute anything

What is encoding?

Input words, get numbers

What is decoding?

Numbers to words

Word2Vec, 100-300

BERT: 768-1024

GPT2: 768-1600

Embeddings can hold a lot more information now

Have meaning of the word in our embeddings

Then we add positional encoding using sin/cos functions

Different sin/cos for each index in embedding vector

Now we have meaning of the word + position

Now we get to attention

Attention did exist before this paper

"The pizza came out of the oven and it tasted good"

We have queries, keys, and values for every token in the sentence, including itself

Matrices we multiply by and the same across each token

Each layer has its own matrices

Layers can be computed in parallel

Difference between each token is only its own embedding and positional encoding

Query:

2D matrix, multiplied by the positional embedding

Meaning of word + position * matrix

Key:

Same thing as query, but instead is multiplied by a key matrix

Value:

Same thing as query, but instead is multiplied by a value matrix

We use queries and keys to calculate similarity between each word and all other words including itself

Advantage of dot product: computational speed b/c addition and multiplication are faster than division

Do softmax to get probabilities

Multiply each value by the probabilities

Now we have our self-attention embedding

Has the proportional amounts of the embeddings of the other words and itself

No longer has itself as its own word

- Add this self-attention embedding back to our word embedding with positional encoder
- Multi-Head attention
 - Each layer specializes in a different thing
 - Can be done in parallel
- Importance of Layers
 - Combines embeddings together of each layer
 - Determine # of layers via trial and error

Discussions:

Make up test

- If you don't take it, +10% to original grade
- Got rid of real world hyperparameter question

Toulmin Argument

- Claim - "dogs nearby"
- Grounds - "I hear dogs barking"
- Warrant - "dogs bark"

- Backing - additional support to warrant
- Qualifier: claim may not be true in all circumstances, has to 'qualify'
- Rebuttal

Text -> Word embeddings -> Positional Encoding -> Self-Attention

Why are embeddings larger?

- Want to eventually have the position and similarities too

Positional Encoding

- Each gets a unique sin/cos function

Self-attention

- Query, Key, Value
 - 1 for each token
 - Query get compared to Key, compared against all other tokens and itself
 - Determines how much of value is used to modify embedding

New Format for Test/Quiz Questions

- Can take make up and write DO NOT GRADE if you just want experience

I made a delicious hawaiian pizza and everyone agreed it tasted good.

- What is the first round of self-attention that the embedding for the token "it" to show strong attention (high Q-K match) to the embedding "delicious Hawaiian pizza"?

- When does self-attention happen?

- N rounds of self-attention, layers

- Claim ("What do I want to prove?"): At least round 3 because to get the full embedding combination of 'delicious Hawaiian pizza' you would have to have been in round 3. It does pair-wise comparison each round.

- Grounds: ("Evidence and specific facts"): Self-attention does pair-wise comparison and compounds the individual token embeddings into a grouped token embedding. In round 1 we have individual tokens, in the 2nd round we have pairs of tokens, then in round 3 we will have 3 token combos.

- Warrant ("Evidence -> Claim"): This means round 3 will be the first time we can compare "it" to "delicious Hawaiian pizza". Since round 3 is the first time that we could have an embedding

representing 'delicious hawaiian pizza', it would not be possible for 'it' to show strong attention to that embedding before round 3.

Backing ("specific example to support"): Multi-round attention gave rise to successful natural language processing.

Qualifier/Acknowledgement: However, no one really understands how these work.

Rebuttal: Alternatively, it may be that after the first round of self-attention that the pairwise blends effectively represents "delicious Hawaiian pizza".

Construction:

Encoder and Decoder only models (BERT/ChatGPT respectively)

Attention only flows in one direction